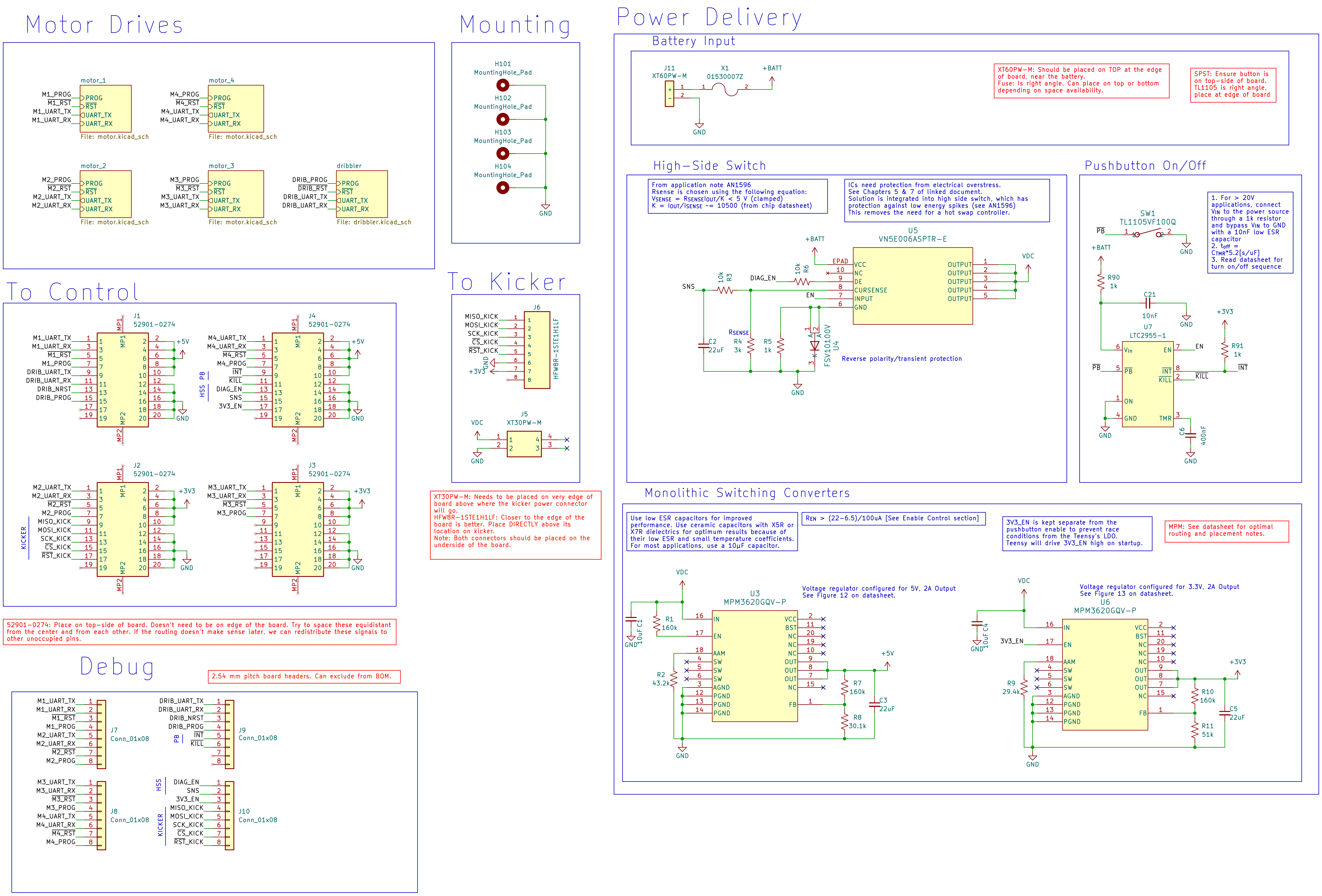
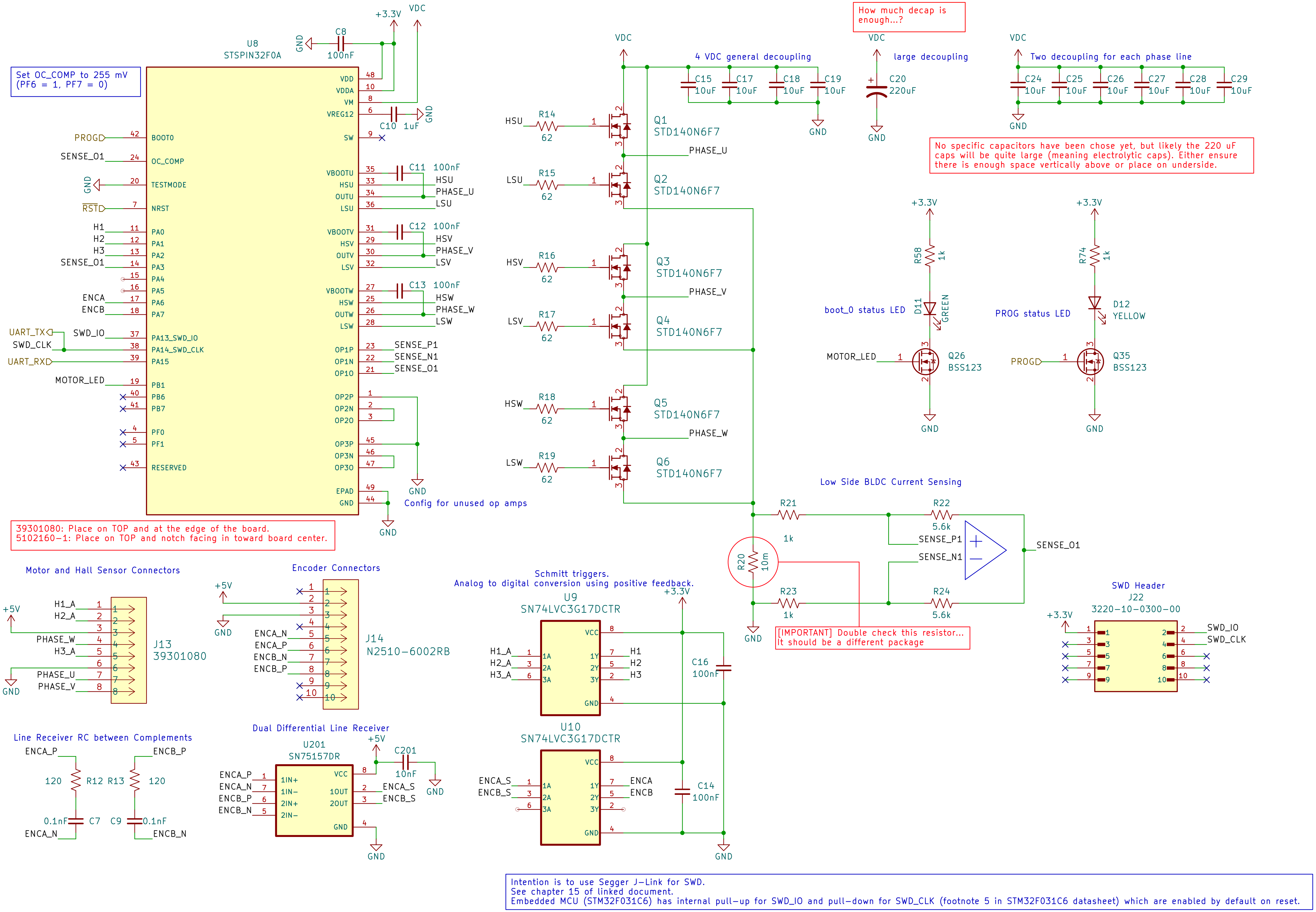
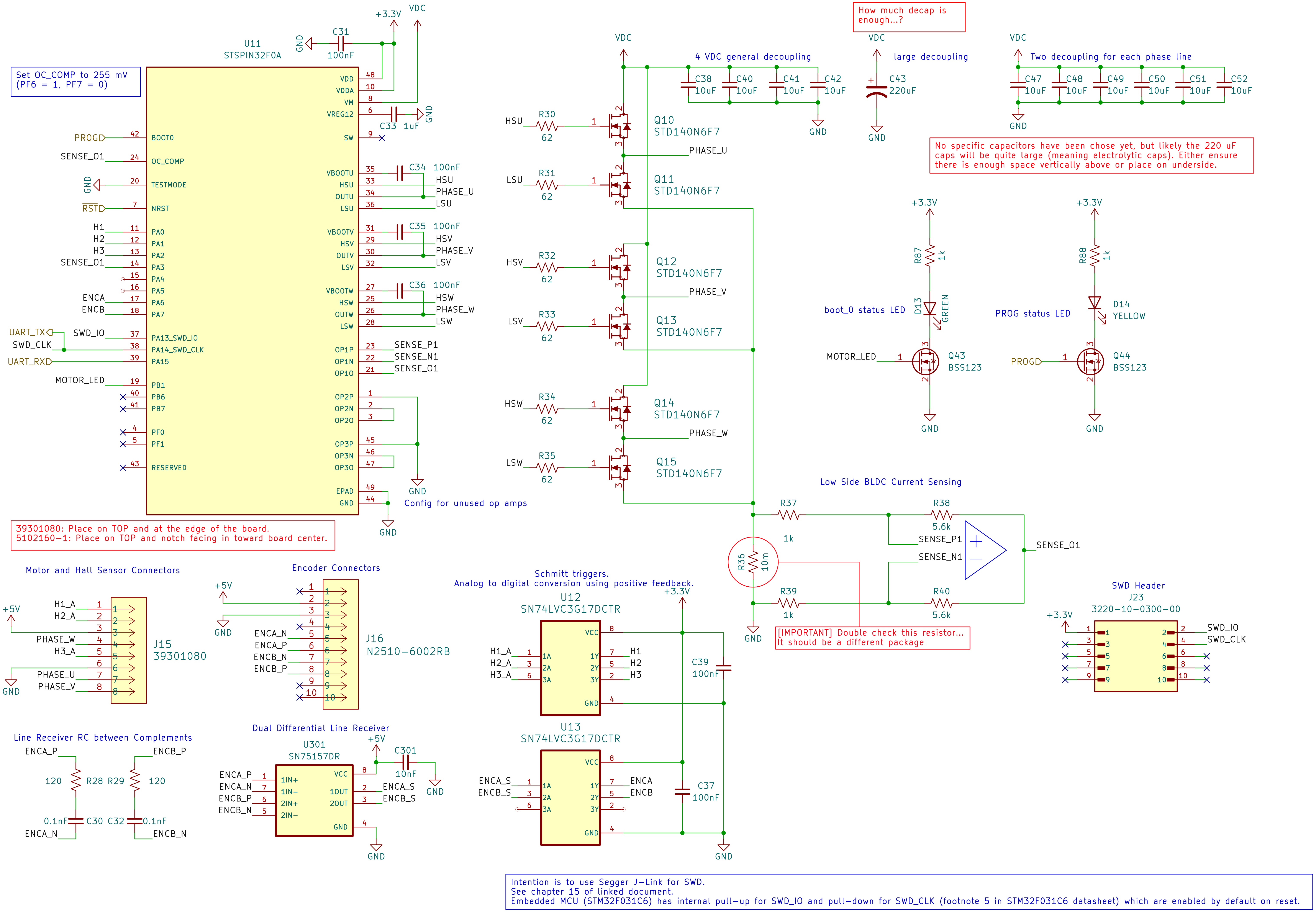
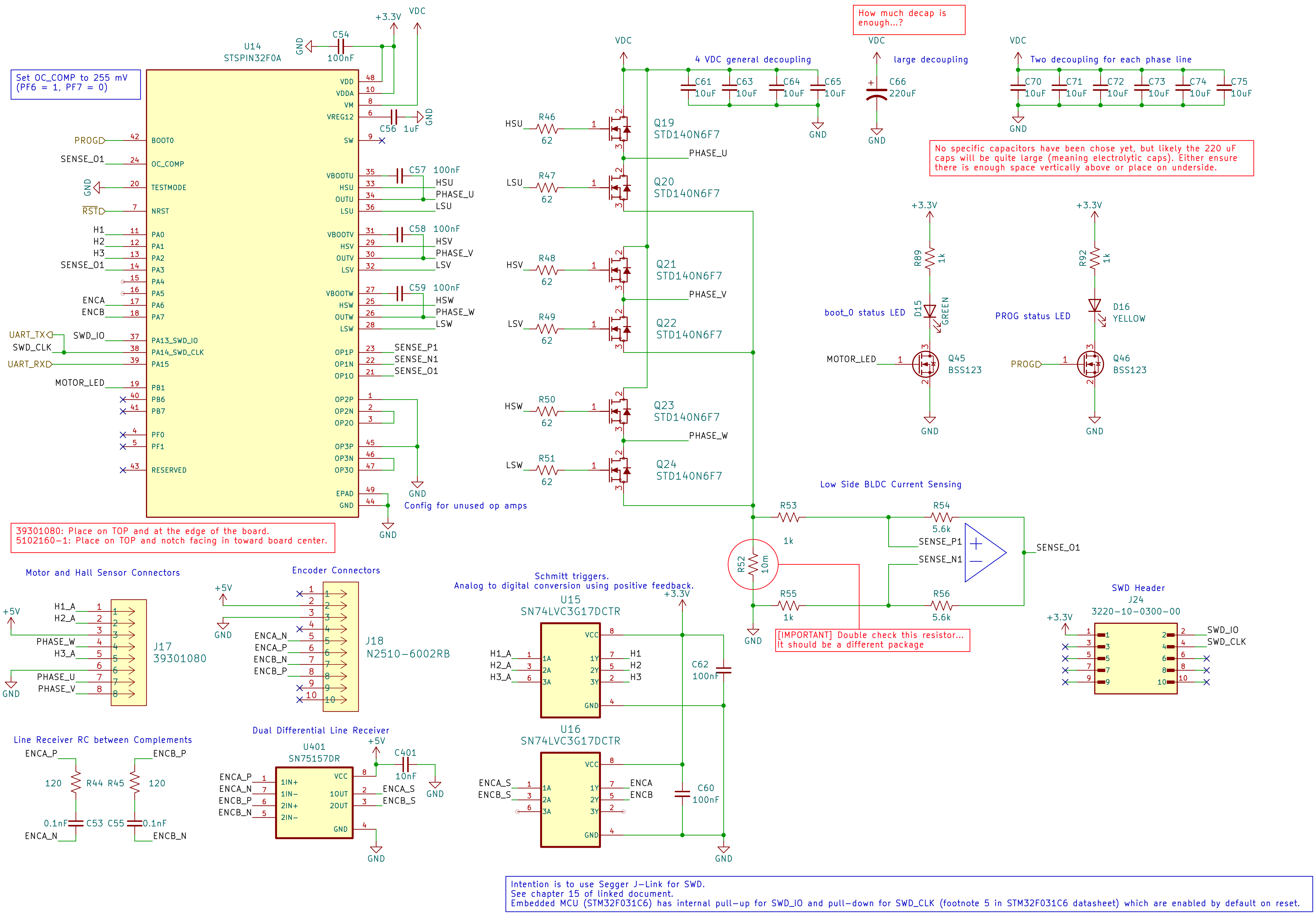
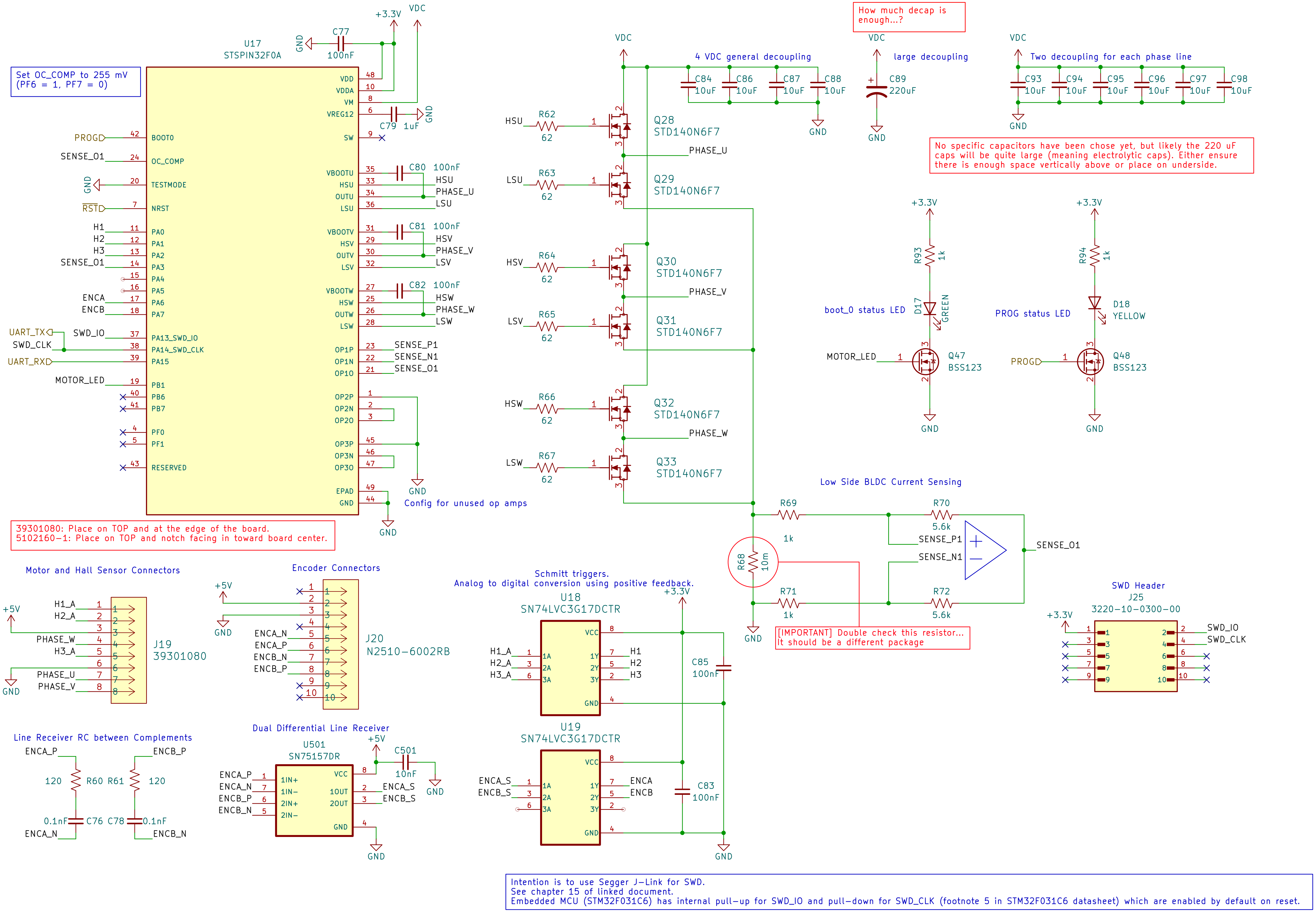

Motor Drives
motor_1
File: motor.kicad_sch
motor_4
File: motor.kicad_sch
motor_2
File: motor.kicad_sch
motor_3
File: motor.kicad_sch
dribbler
File: dribbler.kicad_sch
Mounting
H101 MountingHole_Pad
H102 MountingHole_Pad
H103 MountingHole_Pad
H104 MountingHole_Pad
Power Delivery
Battery Input
J11 XT60PW-M
X1 01530007Z
+BATT
GND
XT60PW-M: Should be placed on TOP at the edge of board, near the battery.
Fuse: Is right angle. Can place on top or bottom depending on space availability.
SPST: Ensure button is on top-side of board. TL1105 is right angle, place at edge of board
To Control
J1 52901-0274
J4 52901-0274
J2 52901-0274
J3 52901-0274
To Kicker
J6 HFW8R-1STE1H1LF
J5 XT30PW-M
XT30PW-M: Needs to be placed on very edge of board above where the kicker power connector will go.
HFW8R-1STE1H1LF: Closer to the edge of the board is better. Place DIRECTLY above its location on kicker.
Note: Both connectors should be placed on the underside of the board.
52901-0274: Place on top-side of board. Doesn't need to be on edge of the board. Try to space these equidistant from the center and from each other. If the routing doesn't make sense later, we can redistribute these signals to other unoccupied pins.
High-Side Switch
From application note AN1596
Rsense is chosen using the following equation:
VSENSE = RSENSEIOUT/K < 5 V (clamped)
K = IOUT/ISENSE ~= 10500 (from chip datasheet)
ICs need protection from electrical overstress.
See Chapters 5 & 7 of linked document.
Solution is integrated into high side switch, which has protection against low energy spikes (see AN1596)
This removes the need for a hot swap controller.
U5 VN5E006ASPTR-E
U4 FSV1010OV
Reverse polarity/transient protection
Pushbutton On/Off
SW1 TL1105VF100Q
U7 LTC2955-1
1. For > 20V applications, connect VIN to the power source through a 1k resistor and bypass VIN to GND with a 10nF low ESR capacitor
2. tOFF = CTMR*5.2[s/uF]
3. Read datasheet for turn on/off sequence
Monolithic Switching Converters
Use low ESR capacitors for improved performance. Use ceramic capacitors with X5R or X7R dielectrics for optimum results because of their low ESR and small temperature coefficients. For most applications, use a 10μF capacitor.
REN > (22-6.5)/100uA [See Enable Control section]
3V3_EN is kept separate from the pushbutton enable to prevent race conditions from the Teensy's LDO. Teensy will drive 3V3_EN high on startup.
MPM: See datasheet for optimal routing and placement notes.
U3 MPM3620GQV-P
Voltage regulator configured for 5V, 2A Output
See Figure 12 on datasheet.
U6 MPM3620GQV-P
Voltage regulator configured for 3.3V, 2A Output
See Figure 13 on datasheet.
Debug
2.54 mm pitch board headers. Can exclude from BOM.
J7 Conn_01x08
J8 Conn_01x08
J9 Conn_01x08
J10 Conn_01x08



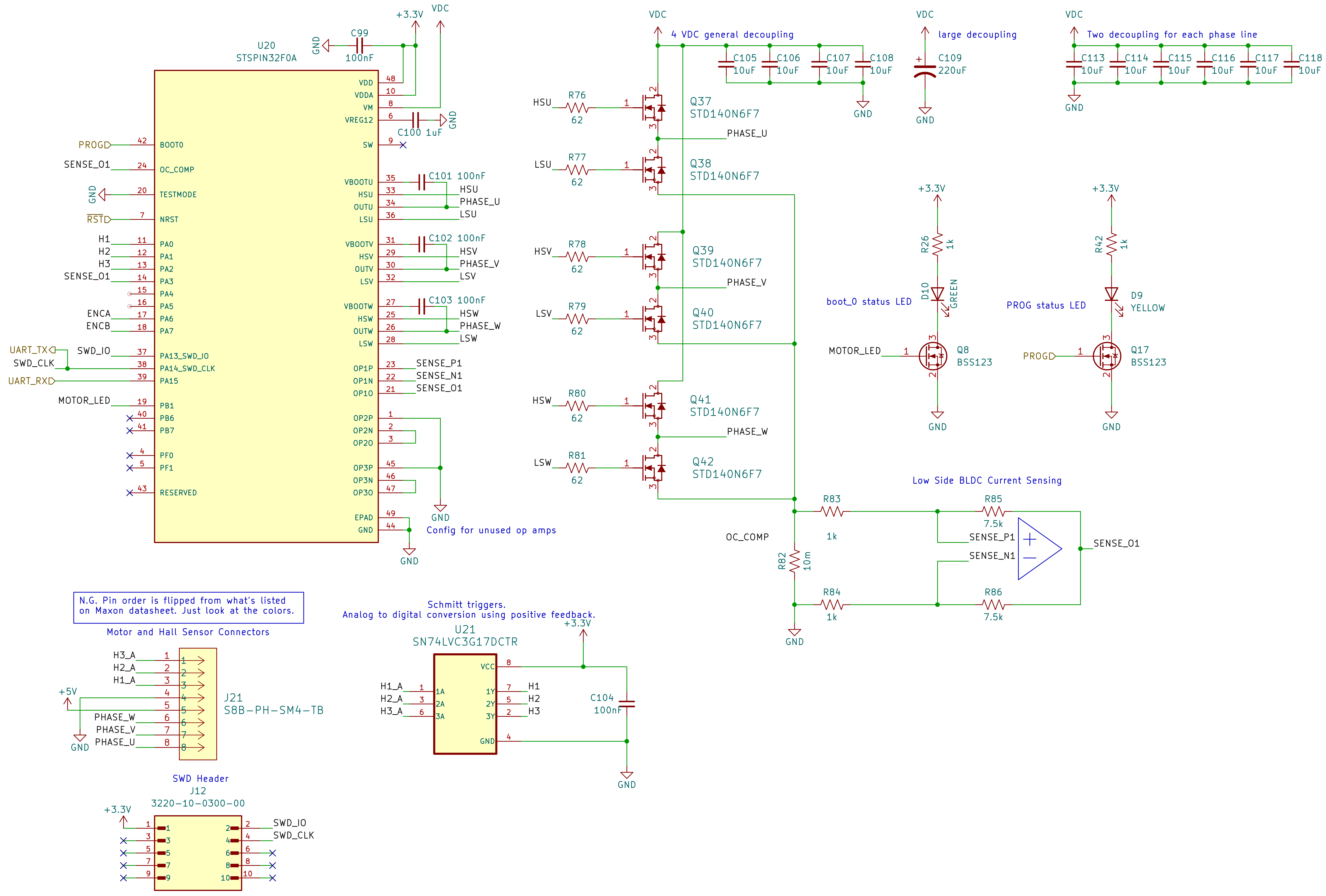
U8
STSPIN32F0A
Set OC_COMP to 255 mV
(PF6 = 1, PF7 = 0)
+3.3V
VDC
GND
C8
100nF
C10 1uF
C11 100nF
C12 100nF
C13 100nF
VDD
VDDA
VM
VREG12
SW
BOOT0
OC_COMP
TESTMODE
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA13_SWD_IO
PA14_SWD_CLK
PA15
PB1
PB6
PB7
PF0
PF1
RESERVED
VBOOTU
HSU
OUTU
LSU
VBOOTV
HSV
OUTV
LSV
VBOOTW
HSW
OUTW
LSW
OP1P
OP1N
OP1O
OP2P
OP2N
OP2O
OP3P
OP3N
OP3O
EPAD
GND
PROG
SENSE_O1
RST
H1
H2
H3
ENCA
ENCB
UART_TX
SWD_IO
SWD_CLK
UART_RXD
MOTOR_LED
PHASE_U
PHASE_V
PHASE_W
SENSE_P1
SENSE_N1
Config for unused op amps
39301080: Place on TOP and at the edge of the board.
5102160-1: Place on TOP and notch facing in toward board center.
R14 62
R15 62
R16 62
R17 62
R18 62
R19 62
Q1 STD140N6F7
Q2 STD140N6F7
Q3 STD140N6F7
Q4 STD140N6F7
Q5 STD140N6F7
Q6 STD140N6F7
4 VDC general decoupling
C15 10uF
C17 10uF
C18 10uF
C19 10uF
How much decap is enough...?
large decoupling
C20 220uF
Two decoupling for each phase line
C24 10uF
C25 10uF
C26 10uF
C27 10uF
C28 10uF
C29 10uF
No specific capacitors have been chose yet, but likely the 220 uF caps will be quite large (meaning electrolytic caps). Either ensure there is enough space vertically above or place on underside.
R58 1k
D11 GREEN
boot_0 status LED
Q26 BSS123
R74 1k
D12 YELLOW
PROG status LED
Q35 BSS123
Low Side BLDC Current Sensing
R21 1k
R22 5.6k
R20 10m
R23 1k
R24 5.6k
[IMPORTANT] Double check this resistor...
It should be a different package
Motor and Hall Sensor Connectors
+5V
H1_A
H2_A
PHASE_W
H3_A
PHASE_U
PHASE_V
J13
39301080
Encoder Connectors
ENCA_N
ENCA_P
ENCB_N
ENCB_P
J14
N2510-6002RB
Schmitt triggers.
Analog to digital conversion using positive feedback.
U9
SN74LVC3G17DCTR
C16 100nF
U10
SN74LVC3G17DCTR
C14 100nF
ENCA_S
ENCB_S
SWD Header
J22
3220-10-0300-00
Line Receiver RC between Complements
120 R12 R13 120
0.1nF C7 C9 0.1nF
Dual Differential Line Receiver
U201
SN75157DR
1IN+
1IN-
2IN+
2IN-
VCC
1OUT
2OUT
C201
10nF
Intention is to use Segger J-Link for SWD.
See chapter 15 of linked document.
Embedded MCU (STM32F031C6) has internal pull-up for SWD_IO and pull-down for SWD_CLK (footnote 5 in STM32F031C6 datasheet) which are enabled by default on reset.


U11
STSPIN32F0A
Set OC_COMP to 255 mV
(PF6 = 1, PF7 = 0)
+3.3V
VDC
GND
C31
100nF
C33 1uF
C34 100nF
C35 100nF
C36 100nF
VDD
VDDA
VM
VREG12
SW
BOOT0
OC_COMP
TESTMODE
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA13_SWD_IO
PA14_SWD_CLK
PA15
PB1
PB6
PB7
PF0
PF1
RESERVED
VBOOTU
HSU
OUTU
LSU
VBOOTV
HSV
OUTV
LSV
VBOOTW
HSW
OUTW
LSW
OP1P
OP1N
OP1O
OP2P
OP2N
OP2O
OP3P
OP3N
OP3O
EPAD
GND
PROG
SENSE_O1
RST
H1
H2
H3
ENCA
ENCB
UART_TX
SWD_IO
SWD_CLK
UART_RX
MOTOR_LED
PHASE_U
PHASE_V
PHASE_W
SENSE_P1
SENSE_N1
Config for unused op amps
39301080: Place on TOP and at the edge of the board.
5102160-1: Place on TOP and notch facing in toward board center.
4 VDC general decoupling
C38 10uF
C40 10uF
C41 10uF
C42 10uF
How much decap is enough...?
large decoupling
C43 220uF
Two decoupling for each phase line
C47 10uF
C48 10uF
C49 10uF
C50 10uF
C51 10uF
C52 10uF
No specific capacitors have been chose yet, but likely the 220 uF caps will be quite large (meaning electrolytic caps). Either ensure there is enough space vertically above or place on underside.
R30 62
R31 62
R32 62
R33 62
R34 62
R35 62
Q10 STD140N6F7
Q11 STD140N6F7
Q12 STD140N6F7
Q13 STD140N6F7
Q14 STD140N6F7
Q15 STD140N6F7
R87 1k
R88 1k
D13 GREEN
D14 YELLOW
boot_0 status LED
PROG status LED
Q43 BSS123
Q44 BSS123
Low Side BLDC Current Sensing
R37 1k
R38 5.6k
R36 10m
R39 1k
R40 5.6k
[IMPORTANT] Double check this resistor...
It should be a different package
Motor and Hall Sensor Connectors
+5V
H1_A
H2_A
H3_A
J15
39301080
Encoder Connectors
ENCA_N
ENCA_P
ENCB_N
ENCB_P
J16
N2510-6002RB
Schmitt triggers.
Analog to digital conversion using positive feedback.
U12
SN74LVC3G17DCTR
U13
SN74LVC3G17DCTR
1A
2A
3A
1Y
2Y
3Y
VCC
C39 100nF
C37 100nF
ENCA_S
ENCB_S
SWD Header
J23
3220-10-0300-00
Line Receiver RC between Complements
120
R28 R29
0.1nF C30 C32 0.1nF
Dual Differential Line Receiver
U301
SN75157DR
1IN+
1IN-
2IN+
2IN-
1OUT
2OUT
C301
10nF
Intention is to use Segger J-Link for SWD.
See chapter 15 of linked document.
Embedded MCU (STM32F031C6) has internal pull-up for SWD_IO and pull-down for SWD_CLK (footnote 5 in STM32F031C6 datasheet) which are enabled by default on reset.


U14
STSPIN32F0A
Set OC_COMP to 255 mV
(PF6 = 1, PF7 = 0)
+3.3V
VDC
GND
C54
100nF
VDD
VDDA
VM
VREG12
C56 1uF
SW
BOOT0
OC_COMP
TESTMODE
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA13_SWD_IO
PA14_SWD_CLK
PA15
PB1
PB6
PB7
PF0
PF1
RESERVED
PROG
SENSE_O1
RST
H1
H2
H3
ENCA
ENCB
UART_TX
SWD_IO
SWD_CLK
UART_RX
MOTOR_LED
VBOOTU
HSU
OUTU
LSU
C57 100nF
PHASE_U
VBOOTV
HSV
OUTV
LSV
C58 100nF
PHASE_V
VBOOTW
HSW
OUTW
LSW
C59 100nF
PHASE_W
OP1P
OP1N
OP1O
SENSE_P1
SENSE_N1
OP2P
OP2N
OP2O
OP3P
OP3N
OP3O
EPAD
Config for unused op amps
39301080: Place on TOP and at the edge of the board.
5102160-1: Place on TOP and notch facing in toward board center.
How much decap is enough...?
4 VDC general decoupling
C61 10uF
C63 10uF
C64 10uF
C65 10uF
large decoupling
C66 220uF
Two decoupling for each phase line
C70 10uF
C71 10uF
C72 10uF
C73 10uF
C74 10uF
C75 10uF
No specific capacitors have been chose yet, but likely the 220 uF caps will be quite large (meaning electrolytic caps). Either ensure there is enough space vertically above or place on underside.
R46 62
R47 62
R48 62
R49 62
R50 62
R51 62
Q19 STD140N6F7
Q20 STD140N6F7
Q21 STD140N6F7
Q22 STD140N6F7
Q23 STD140N6F7
Q24 STD140N6F7
R89 1k
R92 1k
D15 GREEN
D16 YELLOW
boot_0 status LED
PROG status LED
Q45 BSS123
Q46 BSS123
Low Side BLDC Current Sensing
R53 1k
R54 5.6k
R52 10m
R55 1k
R56 5.6k
[IMPORTANT] Double check this resistor...
It should be a different package
Motor and Hall Sensor Connectors
+5V
H1_A
H2_A
PHASE_W
H3_A
PHASE_U
PHASE_V
J17
39301080
Encoder Connectors
ENCA_N
ENCA_P
ENCB_N
ENCB_P
J18
N2510-6002RB
Schmitt triggers.
Analog to digital conversion using positive feedback.
U15
SN74LVC3G17DCTR
C62 100nF
U16
SN74LVC3G17DCTR
ENCA_S
ENCB_S
C60 100nF
SWD Header
J24
3220-10-0300-00
Line Receiver RC between Complements
120 R44 R45 120
0.1nF C53 C55 0.1nF
Dual Differential Line Receiver
U401
SN75157DR
1IN+
1IN-
2IN+
2IN-
1OUT
2OUT
C401
10nF
Intention is to use Segger J-Link for SWD.
See chapter 15 of linked document.
Embedded MCU (STM32F031C6) has internal pull-up for SWD_IO and pull-down for SWD_CLK (footnote 5 in STM32F031C6 datasheet) which are enabled by default on reset.



How much decap is enough...?
VDC
4 VDC general decoupling
large decoupling
Two decoupling for each phase line
C84 10uF
C86 10uF
C87 10uF
C88 10uF
C89 220uF
C93 10uF
C94 10uF
C95 10uF
C96 10uF
C97 10uF
C98 10uF
GND
No specific capacitors have been chose yet, but likely the 220 uF caps will be quite large (meaning electrolytic caps). Either ensure there is enough space vertically above or place on underside.
Set OC_COMP to 255 mV (PF6 = 1, PF7 = 0)
U17
STSPIN32F0A
+3.3V
C77 100nF
C79 1uF
VDD
VDDA
VM
VREG12
SW
BOOT0
OC_COMP
TESTMODE
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA13_SWD_IO
PA14_SWD_CLK
PA15
PB1
PB6
PB7
PF0
PF1
RESERVED
VBOOTU
HSU
OUTU
LSU
VBOOTV
HSV
OUTV
LSV
VBOOTW
HSW
OUTW
LSW
OP1P
OP1N
OP1O
OP2P
OP2N
OP2O
OP3P
OP3N
OP3O
EPAD
GND
PROG
SENSE_O1
RST
H1
H2
H3
ENCA
ENCB
UART_TX
SWD_IO
SWD_CLK
UART_RX
MOTOR_LED
C80 100nF
C81 100nF
C82 100nF
PHASE_U
PHASE_V
PHASE_W
SENSE_P1
SENSE_N1
Config for unused op amps
39301080: Place on TOP and at the edge of the board.
5102160-1: Place on TOP and notch facing in toward board center.
R62 62
R63 62
R64 62
R65 62
R66 62
R67 62
Q28 STD140N6F7
Q29 STD140N6F7
Q30 STD140N6F7
Q31 STD140N6F7
Q32 STD140N6F7
Q33 STD140N6F7
R93 1k
R94 1k
D17 GREEN
D18 YELLOW
boot_0 status LED
PROG status LED
Q47 BSS123
Q48 BSS123
Low Side BLDC Current Sensing
R69 1k
R70 5.6k
R71 1k
R72 5.6k
R68 10m
[IMPORTANT] Double check this resistor...
It should be a different package
Motor and Hall Sensor Connectors
+5V
H1_A
H2_A
PHASE_W
H3_A
PHASE_U
PHASE_V
J19
39301080
Encoder Connectors
ENCA_N
ENCA_P
ENCB_N
ENCB_P
J20
N2510-6002RB
Schmitt triggers.
Analog to digital conversion using positive feedback.
U18
SN74LVC3G17DCTR
U19
SN74LVC3G17DCTR
1A
2A
3A
1Y
2Y
3Y
VCC
C85 100nF
C83 100nF
ENCA_S
ENCB_S
SWD Header
J25
3220-10-0300-00
Line Receiver RC between Complements
R60 120
R61 120
C76 0.1nF
C78 0.1nF
Dual Differential Line Receiver
U501
SN75157DR
1IN+
1IN-
2IN+
2IN-
1OUT
2OUT
C501 10nF
Intention is to use Segger J-Link for SWD.
See chapter 15 of linked document.
Embedded MCU (STM32F031C6) has internal pull-up for SWD_IO and pull-down for SWD_CLK (footnote 5 in STM32F031C6 datasheet) which are enabled by default on reset.


U20
STSPIN32F0A
VDD
VDDA
VM
VREG12
SW
BOOT0
OC_COMP
TESTMODE
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA13_SWD_IO
PA14_SWD_CLK
PA15
PB1
PB6
PB7
PF0
PF1
RESERVED
VBOOTU
HSU
OUTU
LSU
VBOOTV
HSV
OUTV
LSV
VBOOTW
HSW
OUTW
LSW
OP1P
OP1N
OP1O
OP2P
OP2N
OP2O
OP3P
OP3N
OP3O
EPAD
GND
+3.3V
VDC
C99
100nF
C100 1uF
C101 100nF
C102 100nF
C103 100nF
PROG
SENSE_O1
RST
H1
H2
H3
ENCA
ENCB
UART_TX
SWD_IO
SWD_CLK
UART_RX
MOTOR_LED
PHASE_U
PHASE_V
PHASE_W
SENSE_P1
SENSE_N1
Config for unused op amps
4 VDC general decoupling
C105 10uF
C106 10uF
C107 10uF
C108 10uF
large decoupling
C109 220uF
Two decoupling for each phase line
C113 10uF
C114 10uF
C115 10uF
C116 10uF
C117 10uF
C118 10uF
R76 62
R77 62
R78 62
R79 62
R80 62
R81 62
Q37 STD140N6F7
Q38 STD140N6F7
Q39 STD140N6F7
Q40 STD140N6F7
Q41 STD140N6F7
Q42 STD140N6F7
R26 1k
D10 GREEN
boot_0 status LED
Q8 BSS123
R42 1k
D9 YELLOW
PROG status LED
Q17 BSS123
Low Side BLDC Current Sensing
R83 1k
R84 1k
R85 7.5k
R86 7.5k
R82 10m
OC_COMP
N.G. Pin order is flipped from what's listed on Maxon datasheet. Just look at the colors.
Motor and Hall Sensor Connectors
H3_A
H2_A
H1_A
+5V
J21
S8B-PH-SM4-TB
Schmitt triggers.
Analog to digital conversion using positive feedback.
U21
SN74LVC3G17DCTR
VCC
1A
2A
3A
1Y
2Y
3Y
C104
100nF
SWD Header
J12
3220-10-0300-00